

Mia Francis Bonini

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SUMMARY

PhD engineer with broad expertise in computational modeling, medical device evaluation, and clinical research. Experienced building and validating physics-based simulations against real-world experimental and clinical data and leading research studies from IRB submission through publication. Skilled in Python and MATLAB for data analysis and automation, with an extensive record of peer-reviewed publications, technical presentations, and cross-functional collaboration.

SKILLS

Imaging & Ultrasound: ultrasound/echocardiography, CT, MRI, image segmentation and 3D anatomical reconstruction

Clinical Collaboration & Training: direct partnership with cardiac surgeons, interventional radiologists, and clinical staff; case-based technical support; training and mentoring; technical presentations to clinical and multidisciplinary audiences

Cross-Functional Coordination: collaboration across engineers, clinicians, and research teams; feedback loops with R&D on device performance; management of multiple concurrent projects and deadlines

Computer: Microsoft Word, Excel, PowerPoint, Outlook; Python, MATLAB, linear regression, random forest, neural networks

EXPERIENCE

Research Staff | *University of Michigan — Cardio Biomechanics Lab* 11/2025 – present

- Leading development of a physics-based computational modeling framework to simulate complex biological systems and evaluate medical device performance, with a focus on cardiovascular applications.
- Translating clinical and imaging data into quantitative models that inform device design, treatment strategy, and risk assessment ahead of clinical trials.

Graduate Student Research Assistant | *University of Michigan — Cardio Biomechanics Lab* 08/2020 – 11/2025

- Expertise in data driven CFD and FSI modeling.
- Developed patient-specific anatomical models from clinical imaging data (CT, MRI, ultrasound) and validated simulation outputs against clinical measurements, achieving <10% error
- Defined user needs and device performance requirements through direct collaboration with physicians and engineers, translating clinical questions into actionable research and design criteria
- Designed and executed in-silico feasibility studies, concept studies, and technology assessments to support novel medical device development, including a \$415K funded program and an associated patent application
- Built and maintained clinical data pipelines integrating large-scale imaging and electronic health record (EHR) data, applying statistical and machine learning models (Python, Matlab) for outcome prediction
- Contributed to 4 IRB submissions and completed human-subjects research compliance training, ensuring data integrity and regulatory compliance throughout
- Led multiple concurrent research projects to on-time completion; authored 12+ peer-reviewed publications and delivered 7+ international presentations
- Mentored 14 junior researchers in computational methods, data analysis, and scientific communication

Undergraduate Researcher | *Wake Forest University — Dept. of Biomedical Engineering* 06/2019 – 08/2019

- Built an automated pipeline to segment MRI images and analyze muscle fat concentration for a NASA-funded spaceflight study, while maintaining HIPAA-compliant datasets.
- Extracted spinal bone and muscle tissue and performed mechanical compression testing to characterize tissue properties.

Engineering Intern — SSD & Thermal Solutions | *Intel Corporation* 06/2018 – 08/2018

- Characterized thermal and mechanical performance of device assemblies using wind tunnel and still-air chamber testing, analyzing temperature profiles.
- Standardized laboratory testing protocols and documentation to reduce operator variability and improve reproducibility.

EDUCATION

Ph.D. in Biomedical Engineering — University of Michigan 11/2025

M.S. in Biomedical Engineering — University of Michigan 05/2022

B.S. in Mechanical Engineering — University of Portland 05/2020

SELECTED PUBLICATIONS

1. **Bonini M**, et al. A monolithic patient-specific 3D–0D model for In silico investigation of hemodynamics in patients with left ventricular assist devices. *Biomechanics and Modeling in Mechanobiology* 25, 51 (2026). <https://doi.org/10.1007/s10237-026-02063-9>
2. **Bonini M**, et al. Image-Based, Whole-System Hemodynamic Modeling of Mitral Regurgitation and Its Impact on the Right Ventricular Function. *Frontiers in Cardiovascular Medicine*; 2026. <https://doi.org/10.3389/fcvm.2025.1673443>
3. **Bonini M**, Hirschvogel M, et al. Patient-Specific Hemodynamic Modeling for Patients with Left Ventricle Assist Device. In: *Functional Imaging and Modeling of the Heart. Lecture Notes in Computer Science*; 2025. https://doi.org/10.1007/978-3-031-94562-5_29
4. **Bonini M**, Sanjay S, et al. Modeling the Hemodynamic Impact of Y-incision Aortic Annular Enlargements on Aortic Valve Replacement and Valve-in-Valve Procedures. *Journal of Cardiovascular and Translational Research*. 2025;18:876–887. <https://doi.org/10.1007/s12265-025-10634-x>
5. **Bonini M**, et al. Modeling the Hemodynamic Impact of Mitral and Aortic Valve Repair in Patients with Left Ventricular Assist Devices. *Computers in Biology and Medicine*. 2026. <https://doi.org/10.48550/arXiv.2609.26939>, (In review).
6. **Bonini M**, et al. Image-Based, Fluid-Structure Interaction Modeling Framework for Aortic Dissection Digital Twin. *Medical Image Analysis*. 2026. (In submission).
7. Sayut JP, Jilberto J, **Bonini M**, Hirschvogel M, Zhang W, Nordsletten DA. In Silico Evaluation of Cardiac Tissue-Engineered Patch Interventions. *Computers in Biology and Medicine*. 2025;198(Pt B):111249. <https://doi.org/10.1016/j.compbiomed.2025.111249>
8. Hirschvogel M, **Bonini M**, Balmus M, Nordsletten D. Effective Block Preconditioners for Fluid Dynamics Coupled to Reduced Models of a Non-Local Nature. *Computer Methods in Applied Mechanics and Engineering*. 2025;435:117541. doi:[10.1016/j.cma.2024.117541](https://doi.org/10.1016/j.cma.2024.117541)
9. Hirschvogel M, Balmus M, **Bonini M**, Nordsletten D. Fluid-reduced-solid interaction (FrSI): Physics- and projection-based model reduction for cardiovascular applications. *Journal of Computational Physics*. 2024;506:112921. doi:[10.1016/j.jcp.2024.112921](https://doi.org/10.1016/j.jcp.2024.112921)

BOOK CHAPTERS

1. **Bonini M**, Jilberto J, Hirschvogel M, Nordsletten D. (2027) *Biomechanics of the Heart: From Model to Patient*. In Bovendeerd, P., Clarysse, P., Croisille, P., Nash, M. (Eds.), *Biomechanics of Living Organs*. (In preprint)
2. Miller R, Marlevi D, Zhang W, Hirschvogel M, Hadjicharalambous M, Capilnasiu A, Balmus M, Hager S, Jilberto J, **Bonini M**, Wittgenstein A, Ahmed Y, & Nordsletten D. (2021). *Modeling Biomechanics in the Healthy and Diseased Heart*. In J. Málek & E. Süli (Eds.), *Modeling Biomaterials* (pp. 141–239). Springer International Publishing.

SELECTED PRESENTATIONS

3. 1. Mia Bonini, Patient-Specific Hemodynamic Modeling for Patients with Left Ventricle Assist Device. *Functional Imaging and Modeling of the Heart (FIMH)*. June 2025. Dallas, TX. (podium)
4. 2. Mia Bonini, Modeling the Hemodynamic Impact of Aortic Root Enlargements in Aortic Valve Replacement. *WCCM-PANACM*. July 2024. Vancouver, Canada. (podium)
5. 3. Mia Bonini, Modeling the Hemodynamic Impact of Aortic Root Enlargements in Aortic Valve Replacement. *ECCOMAS*. June 2024. Lisbon, Portugal. (podium)
6. 4. Mia Bonini, Image-Based, Whole System Hemodynamic Modeling for Valve Heart Diseases. *22nd Computational Fluids Conference*. April 2023. Cannes, France. (podium)
7. 5. Mia Bonini, Hemodynamic Modeling for Mitral Regurgitation. *9th World Congress of Biomechanics*. July 2022. Taipei, Taiwan. (podium)
8. 6. Mia Bonini, Patient Specific Hemodynamic Modeling of Transcatheter Aortic Valve Replacement. *9th World Congress of Biomechanics*. July 2022. Taipei, Taiwan. (podium)
9. 7. Mia Bonini, Hemodynamic Modeling for Mitral Regurgitation. *11th European Solid Mechanics Conference*. July 2022. Galway, Ireland. (podium)

HONORS & AWARDS

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| 1. Functional Imaging and Modeling of the Heart Early-Career Scholarship Award | 06/2025 |
| 2. Marvin Kirsh Resident Research Symposium, 1st Place Basic Science | 03/2025 |
| 3. University of Michigan Rackham Graduate Student Research Grant | 11/2023 |
| 4. EUROMECH Young Researcher Award | 07/2022 |

LEADERSHIP AND SERVICE

- Trained and mentored 14 undergraduate and MS students in all stages of computational modeling, testing, and experimental research, enhancing technical proficiency and research capabilities.

- Guided 3 students in preparing and presenting research at a national engineering conference, increasing their professional exposure.
- Led a student through the process of writing a research paper, resulting in a co-authored publication.
- Mentored two prospective PhD students from non-traditional paths or historically disadvantaged backgrounds during the PhD application process.